

Variable Selection and Minimax Prediction in High-dimensional Functional Linear Model

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- [Xingche Guo, Yehua Li, and Tailen Hsing](#). “Variable Selection and Minimax Prediction in High-dimensional Functional Linear Model.” arXiv preprint *arXiv:2310.14419* (2023).

Functional linear regression

Consider the functional linear regression model (FLM):

$$Y_i = \sum_{j=1}^p \langle X_{ij}, \beta_{0j} \rangle_2 + \varepsilon_i, \quad i = 1, \dots, n$$

- $X_{ij}(\cdot)$ are functional predictors in $\mathbb{L}_2 := \mathbb{L}_2[0, 1]$ with mean zero
- $\beta_{0j}(\cdot)$ are unknown coefficient (i.e., slope) functions in $\mathbb{L}_2$
- $\langle \cdot, \cdot \rangle_2$ is inner product in $\mathbb{L}_2$
- ε_i are errors in $\mathbb{R}$
- p may be large relative to n

The p -fold product space $\mathbb{L}_2^p$ contains elements $\mathbf{f} = (f_1, \dots, f_p)^\top$, where each $f_j \in \mathbb{L}_2$, and

$$\langle \mathbf{f}, \mathbf{g} \rangle_2 := \sum_{j=1}^p \langle f_j, g_j \rangle_2, \quad \mathbf{f}, \mathbf{g} \in \mathbb{L}_2^p.$$

Notation and assumptions

- $\boldsymbol{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_n)^\top \in \mathbb{R}^n$, where the $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$.
- $\boldsymbol{\beta}_0 = (\beta_{01}, \dots, \beta_{0p})^\top \in \mathbb{L}_2^p$
- $\mathbf{X}_i = (X_{i1}, \dots, X_{ip})^\top \in \mathbb{L}_2^p$, where the $\mathbf{X}_i \stackrel{iid}{\sim} N(0, \mathcal{C})$.
- $\mathcal{C} = \mathbb{E}(\mathbf{X}_1 \otimes \mathbf{X}_1) = \left\{ \mathcal{C}^{(j,j')} \right\}_{j,j'=1}^p = \{\mathbb{E}(X_{1j} \otimes X_{1j'})\}_{j,j'=1}^p$
- $\mathcal{C}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i \otimes \mathbf{X}_i^\top$

Note that $\otimes$ denotes outer product in $\mathbb{L}_2^p$, and $\mathcal{C}$ and $\mathcal{C}_n$ are, respectively, the theoretical and sample covariance operators for the predictor.

Thus, we have

$$Y_i = \langle \mathbf{X}_i, \boldsymbol{\beta}_0 \rangle_2 + \varepsilon_i, \quad i = 1, \dots, n.$$

Connection between FLR and Group Lasso

- Let the j -th functional predictors and the associated functional coefficient expressed as linear combinations of orthonormal basis

$$X_{ij}(t) = \sum_{k=1}^{S_j} \theta_{ijk} \phi_{jk}(t), \quad \beta_j(t) = \sum_{k=1}^{S_j} \eta_{jk} \phi_{jk}(t), \quad \langle \phi_{jk}, \phi_{jl} \rangle_2 = \delta_{kl}.$$

- The FLR can be reformulated as

$$Y_i = \sum_{j=1}^p \sum_{k=1}^{S_j} \theta_{ijk} \eta_{jk} + \epsilon_i.$$

- Variable selection problem for the FLR can be regarded as a group lasso type problem.
- However:
 - 1 We need $S_j \rightarrow \infty$.
 - 2 Decompose $X_{ij}(t)$ and $\beta_j(t)$ using a common basis set might not be the most efficient way.

Variable Selection

Variable selection

Let

$$\mathcal{S} = \{j \in \{1, \dots, p\} : \text{Var}\langle X_{1j}, \beta_{0j} \rangle_2 > 0\}$$

denote the true signal set, and let $q := |\mathcal{S}|$.

Also, define the signal set containing “important” predictors:

$$\mathcal{S}_G := \left\{ j \in \mathcal{S} : \sqrt{\text{Var}\langle X_{1j}, \beta_{0j} \rangle_2} > G \right\},$$

where $G \in (0, \infty)$.

Assuming that $\mathcal{S}$ is sparse, our goal is to consider a procedure that leads to the identification of $\mathcal{S}_G$ or $\mathcal{S}$.

Not much is known about this variable selection problem in the infinite-dimensional case.

A common approach for functional linear regression is based on penalized least squares, namely, estimate $\beta_0 = (\beta_{01}, \dots, \beta_{0p})^\top$ by

$$\hat{\beta} = \operatorname{argmin}_{\beta \in \mathbb{H}} \left\{ \frac{1}{2n} \sum_{i=1}^n (Y_i - \langle \mathbf{X}_i, \beta \rangle_2)^2 + \sum_{j=1}^p \operatorname{Pen}(\beta_j; \boldsymbol{\lambda}) \right\}$$

- $\mathbb{H} \subset \mathbb{L}_2^p$
- $\operatorname{Pen}(\beta_j; \boldsymbol{\lambda})$ is a suitable penalty that will allow us to address both the noninvertibility of the covariance operators $\mathcal{C}_n$ and the goal of achieving sparsity in the estimated signal set.

Our approach assumes that $\mathbb{H} := \otimes_{j=1}^p \mathbb{H}_j$, which is product space of the RKHSs $\mathbb{H}_j$, and adopts an elastic-net type penalty to be specified below.

Reproducing kernel Hilbert space (RKHS)

Recall that a real, symmetric, square-integrable, and nonnegative definite function $K(\cdot, \cdot)$ on $[0, 1]^2$ is called a reproducing kernel for a Hilbert space of functions $\mathbb{H}(K)$ on $[0, 1]$ if

- $K(\cdot, t) \in \mathbb{H}(K)$ for any $t \in [0, 1]$, and
- $\mathbb{H}(K)$ is equipped with the inner product such that

$$\langle \beta, K(\cdot, t) \rangle_{\mathbb{H}(K)} = \beta(t), \quad \beta \in \mathbb{H}(K), t \in [0, 1].$$

The Hilbert space $\mathbb{H}(K)$ is then called an RKHS (Aronszajn, 1950).

The RKHS is a useful framework in nonparametric estimation (Wahba, 1990) and functional data analysis (Cai and Yuan, 2012; Hsing and Eubank, 2015; Sun et al., 2018; Lee et al., 2023). Commonly adopted spaces include Sobolev spaces.

For any reproducing kernel $K(\cdot, \cdot)$, define the integral operator

$$(\mathcal{L}_K f)(\cdot) = \int_0^1 K(s, \cdot) f(s) ds, \quad f \in \mathbb{L}_2,$$

and the square-root integral operator, $\mathcal{L}_{K^{1/2}} := \sqrt{\mathcal{L}_K}$.

By Wahba (1990),

$$\mathcal{L}_{K^{1/2}} : \mathbb{L}_2 \mapsto \mathbb{H}(K)$$

is surjective and isometric, which implies that for all $\beta \in \mathbb{H}(K)$, there exists a unique $f \in \mathbb{L}_2$ such that $\beta = \mathcal{L}_{K^{1/2}} f$ with $\|\beta\|_{\mathbb{H}(K)} = \|f\|_2$.

In our problem, representing the β_j in terms of the corresponding functions f_j leads to simplifications.

Recall we assume $\mathbb{H} := \otimes_{j=1}^p \mathbb{H}_j$, and our penalized least squares estimate is

$$\hat{\beta} = \operatorname{argmin}_{\beta \in \mathbb{H}} \left\{ \frac{1}{2n} \sum_{i=1}^n (y_i - \langle \mathbf{X}_i, \beta \rangle_2)^2 + \sum_{j=1}^p \operatorname{Pen}(\beta_j; \lambda) \right\}.$$

Let

$$\begin{aligned} \beta &= \mathcal{L}_{K^{1/2}} \mathbf{f} := \operatorname{diag} \left(\mathcal{L}_{K_1^{1/2}}, \dots, \mathcal{L}_{K_p^{1/2}} \right) (f_1, \dots, f_p)^\top \\ \widetilde{\mathbf{X}}_i &:= \left(\widetilde{X}_{i1}, \dots, \widetilde{X}_{ip} \right)^\top = \mathcal{L}_{K^{1/2}} \mathbf{X}_i \\ \widetilde{\mathbf{X}}_{[n]} &:= \left(\widetilde{\mathbf{X}}_1, \dots, \widetilde{\mathbf{X}}_n \right)^\top = \mathcal{L}_{K^{1/2}} \mathbf{X}_{[n]}. \end{aligned}$$

Then, $\hat{\beta} = \mathcal{L}_{K^{1/2}} \hat{\mathbf{f}}$, where

$$\hat{\mathbf{f}} = \operatorname{argmin}_{\mathbf{f} \in \mathbb{L}_2^p} \left\{ \frac{1}{2n} \sum_{i=1}^n \left(y_i - \left\langle \widetilde{\mathbf{X}}_i, \mathbf{f} \right\rangle_2 \right)^2 + \sum_{j=1}^p \operatorname{Pen}(f_j; \lambda) \right\}. \quad (*)$$

Functional elastic-net type penalty

We propose to use the following functional elastic-net type penalty

$$\sum_{j=1}^p \text{Pen}(f_j; \lambda_1, \lambda_2) = \lambda_1 \sum_{j=1}^p \|\Psi_j f_j\|_2 + \frac{\lambda_2}{2} \sum_{j=1}^p \|f_j\|_2^2$$

where Ψ_j is an operator on $\mathbb{L}_2$ satisfying the following condition **C.1**.

C.1 For $j = 1, \dots, p$, Ψ_j is a self-adjoint, such that

- ◇ $\Psi_j(\mathbb{M}_{nj}) \subset \mathbb{M}_{nj}$, where $\mathbb{M}_{nj} = \text{Span}\{\tilde{X}_{ij}(\cdot), i = 1, \dots, n\}$;
- ◇ there exist positive constants $0 < C_{\min} < C_{\max} < \infty$ such that, uniformly for all j , the eigenvalues of Ψ_j are in the interval $[C_{\min}, C_{\max}]$.

Representer Theorem: The optimization problem in (*) is convex and, under **C.1**, any minimizer $\hat{f}_j$ of (*) must be in the finite-dimensional space $\mathbb{M}_{nj}$.

More notations

Theoretical covariance operators

- $\mathcal{T} := \mathbb{E} \left(\widetilde{\mathbf{X}}_1 \otimes \widetilde{\mathbf{X}}_1 \right) = \mathcal{L}_{\mathbf{K}^{1/2}} \mathcal{C} \mathcal{L}_{\mathbf{K}^{1/2}} =: \{\mathcal{T}^{(j,j')}\}_{j,j'=1}^p$
- $\mathcal{T}^{(\mathcal{S},\mathcal{S})} := \{\mathcal{T}^{(j,j')}\}_{j,j' \in \mathcal{S}}, \quad \rho_1 := \text{max eigenvalue of } \mathcal{T}^{(\mathcal{S},\mathcal{S})}$
- $\mathcal{Q}^{(\mathcal{S},\mathcal{S})} := \text{diag}\{\mathcal{T}^{(j,j)}, j \in \mathcal{S}\}$
- $\mathcal{T}^{(\mathcal{S}^c,\mathcal{S})} := \{\mathcal{T}^{(j,j')}\}_{j \in \mathcal{S}^c, j' \in \mathcal{S}}$

Sample covariance operators

- $\mathcal{T}_n := n^{-1} \widetilde{\mathbf{X}}_{[n]}^\top \otimes \widetilde{\mathbf{X}}_{[n]} = \mathcal{L}_{\mathbf{K}^{1/2}} \mathcal{C}_n \mathcal{L}_{\mathbf{K}^{1/2}} =: \{\mathcal{T}_n^{(j,j')}\}_{j,j'=1}^p$

$$\sum_{j=1}^p \text{Pen}(f_j; \lambda_1, \lambda_2) = \lambda_1 \sum_{j=1}^p \|\Psi_j f_j\|_2 + \frac{\lambda_2}{2} \sum_{j=1}^p \|f_j\|_2^2$$

- The $\mathbb{L}_2$ -norm $\|f_j\|_2$ in $\text{Pen}(f_j; \lambda_1, \lambda_2)$ corresponds to the RKHS norm $\|\beta_j\|_{\mathbb{H}_j}$, a commonly used norm in functional regression problems; see, e.g., Crambes et al. (2009); Cai and Yuan (2012). Both papers discussed the optimality property of the approach.

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- A simple choice for Ψ_j is $\Psi_j = I$, the identity operator, based on which the combined penalty $\text{Pen}(f_j; \lambda_1, \lambda_2)$ includes both $\|f_j\|_2$ and $\|f_j\|_2^2$ and resembles the group elastic-net penalty (Zou and Hastie, 2005; Yuan and Lin, 2006). Then $C_{\min} = C_{\max} = 1$.

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- To achieve better prediction performance, Xue and Yao (2021) considered an L_1 penalty that focused on the **amount of variation** that X_j explains rather than the norm of f_j . Their penalty translates in our setting to

$$\|\Psi_j f_j\|_2 = \widehat{\text{Var}}^{1/2} \langle X_{1j}, \beta_j \rangle_2,$$

namely, $\Psi_j = \sqrt{\mathcal{T}_n^{(j,j)}}$, with $\mathcal{T}_n^{(j,j)}$ = the sample covariance operator of the j -th predictor, or the (j, j) -th block of $\mathcal{T}_n$.

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- The above Ψ_j does not satisfy [C.1](#), one solution is to let

$$\Psi_j = \sqrt{\mathcal{T}_n^{(j,j)} + \theta \mathcal{I}}.$$

Functional KKT condition

For the classical lasso problem, the Karush-Kuhn-Tucker (KKT) condition can be used to characterize the solution (e.g., Zhao and Yu, 2006; Wainwright, 2009), where subgradients are used in place of gradients due to the nondifferentiability of the lasso objective function.

Similarly, in the functional-data setting, the objective function of (*) is not always differentiable due to the group-lasso-type penalty involving $\|\Psi_j f_j\|_2$.

Let f_0 be the true function of f . Using the notion of Gateaux sub-differential, the solution $\hat{f}$ for (*) exists uniquely and satisfies the **functional KKT** condition:

$$\mathcal{J}_n(\hat{f} - f_0) - n^{-1} \widetilde{\mathbf{X}}_{[n]}^\top \boldsymbol{\varepsilon}_n + \lambda_2 \hat{f} + \lambda_1 \boldsymbol{\omega} = 0,$$

where

$$\omega_j = \begin{cases} \frac{\Psi_j^2 \hat{f}_j}{\|\Psi_j \hat{f}_j\|_2} & \text{if } \hat{f}_j \neq 0 \\ \Psi_j \eta_j & \text{for some } \eta_j \text{ with } \|\eta_j\|_2 \leq 1 \text{ if } \hat{f}_j = 0. \end{cases}$$

Some conditions

C.1 For $j = 1, \dots, p$, Ψ_j is a self-adjoint operator such that

- ◇ $\Psi_j(\mathbb{M}_{nj}) \subset \mathbb{M}_{nj}$, where $\mathbb{M}_{nj} = \text{Span}\{\tilde{X}_{ij}(\cdot), i = 1, \dots, n\}$;
- ◇ there exist positive constants $0 < C_{\min} < C_{\max} < \infty$ such that, uniformly for all j , the eigenvalues of Ψ_j are in the interval $[C_{\min}, C_{\max}]$.

C.2 Each $\mathcal{T}^{(j,j)}$ is standardized such that $\|\mathcal{T}^{(j,j)}\|_2 = 1$, with its trace uniformly bounded by a finite constant τ .

For any $\mathbf{f} \in \mathbb{L}_2^p$, $\|\mathbf{f}\|_\infty := \max_{1 \leq j \leq p} \|f_j\|_2$.

For any $p \times p$ operator-valued matrix $\mathcal{A} = (\mathcal{A}_{ij})_{i,j=1}^p$ that maps from $\mathbb{L}_2^p$ to $\mathbb{L}_2^p$ (each $\mathcal{A}_{ij}$ maps from $\mathbb{L}_2$ to $\mathbb{L}_2$), define the norm

$$\|\mathcal{A}\|_{\infty, \infty} := \sup_{\|\mathbf{f}\|_\infty \leq 1} \|\mathcal{A}\mathbf{f}\|_\infty.$$

C.3 For some $\gamma \in (0, 1]$,

$$\kappa(\lambda_2) \cdot \left\| \mathcal{J}^{(\mathcal{S}^c, \mathcal{S})}(\mathcal{J}^{(\mathcal{S}, \mathcal{S})})^{-1} \right\|_{\infty, \infty} \leq (C_{\min}/C_{\max})(1 - \gamma),$$

$$\text{where } \kappa(\lambda_2) := \left\| \mathcal{J}^{(\mathcal{S}, \mathcal{S})}(\mathcal{J}^{(\mathcal{S}, \mathcal{S})} + \lambda_2 I)^{-1} \right\|_{\infty, \infty}.$$

Condition **C.3** controls the correlation between functional predictors in the true signal set $\mathcal{S}$ and those in the non-signal set $\mathcal{S}^c$. This assumption is related to the so-called *irrepresentable condition* on model selection consistency of the classical lasso (Zhao and Yu, 2006; Wainwright, 2009), the classical elastic-net (Jia and Yu, 2010), and the sparse additive models (Ravikumar et al., 2009).

Condition **C.3** becomes harder to fulfill when $\kappa(\lambda_2)$ or $C_{\max}/C_{\min}$ is large. However, when the predictors in $\mathcal{S}$ and in $\mathcal{S}^c$ are uncorrelated, then $\left\| \mathcal{J}^{(\mathcal{S}^c, \mathcal{S})}(\mathcal{J}^{(\mathcal{S}, \mathcal{S})})^{-1} \right\|_{\infty, \infty} = 0$ and the assumption holds trivially.

$$\text{C.4 } \aleph(\lambda_2) := \left\| \left(\mathcal{T}^{(\mathcal{S}, \mathcal{S})} - \mathcal{Q}^{(\mathcal{S}, \mathcal{S})} \right) \left(\mathcal{Q}^{(\mathcal{S}, \mathcal{S})} + \lambda_2 I \right)^{-1} \right\|_{\infty, \infty} < 1.$$

Condition **C.4** puts constraints on the correlations between the predictors in the true signal set $\mathcal{S}$, so that none of the true predictors can be represented by other predictors in $\mathcal{S}$.

When the predictors in $\mathcal{S}$ are uncorrelated, then $\aleph(\lambda_2) = 0$ and **C.4** trivially holds.

Restrictions on λ_1, λ_2

Define the following conditions on λ_1, λ_2 :

- $\lambda_1/\lambda_2 > \left(\frac{3}{\gamma} - 2\right) C_{\max}^{-1}$
- $D_{1,1}^* > \lambda_1 > D_{1,2}^* \frac{\tau^{1/2}(1+\sigma)}{C_{\min}\gamma} \sqrt{\frac{\log(p-q)}{n}}$
- $D_{2,1}^* > \lambda_2 > D_{2,2}^* \frac{\tau(1+\sigma)(\rho_1+1)}{(C_{\min}/C_{\max})^2\gamma^2} \max\left(\frac{q \log(p-q)}{n}, \sqrt{\frac{q^2}{n}}\right)$

where $D_{1,1}^*, D_{1,2}^*, D_{2,1}^*, D_{2,2}^*$ are universal constants.

By carefully separating the model/regularization parameters with universal constants, our nonasymptotic results below can be readily used to state asymptotic results where some or all of the parameters could change with n .

Variable selection consistency

Theorem 1

Under the conditions stated, $\hat{S} := \{j : \hat{f}_j \neq 0\}$ exists uniquely, and (i) and (ii) below hold with probability at least

$$1 - \exp\left(-D \frac{\lambda_2^2 n}{q}\right), \quad \text{where} \quad D = D^* \left(\frac{(C_{\min}/C_{\max})\gamma}{\tau^{1/2}(\rho_1 + 1)(\sigma + 1)} \right)^2,$$

for some universal constant D^ .*

- (i) $\hat{S} \subset S$, i.e., only true predictors are selected by the procedure.
- (ii) $\hat{S} \supset S_G$ for

$$G = \frac{12 - 8\aleph(\lambda_2)}{1 - \aleph(\lambda_2)} \left(C_{\max} \sqrt{\lambda_1^2/\lambda_2} + 2\sqrt{\lambda_2} \right);$$

in particular, if $S_G = S$, then $\hat{S} = S$ and variable selection consistency is achieved.

Minimax Prediction

Excess prediction risk

Consider the excess prediction risk

$$\mathcal{R}(\mathbf{f}) := \mathbb{E} \left\langle \widetilde{\mathbf{X}}_{1,j}, \mathbf{f}_0 - \mathbf{f} \right\rangle_2^2$$

as a metric to measure prediction accuracy.

Theorem 2

The excess prediction risk based on our estimator, $\hat{\mathbf{f}}$, satisfies

$$\mathcal{R}(\hat{\mathbf{f}}) < q \left(4C_{\max} \lambda_1 + 4\lambda_2 + C_{\max}^2 \lambda_1^2 / \lambda_2 \right)$$

with probability bounded below by the expression in the previous result.

Up to now, the bounds presented are non-asymptotic. We can easily state asymptotic results by allowing p, q as well as the model/regularization parameters to vary with the sample size n .

Corollary 1

Assume that $C_{\min}$ and γ are bounded away from 0, and ρ_1, σ^2, τ , and $C_{\max}$ bounded away from ∞ . Let

$$\alpha(p, q, n) := q \max \left(q, \sqrt{\log(p - q)}, \sqrt{q \log n} \right)$$

and assume that $\alpha(p, q, n) = o(n^{1/2})$.

Then, with appropriate λ_1, λ_2 and some sufficiently large constant D ,

$$\mathbb{P} \left(\mathcal{R}(\hat{\mathbf{f}}) > Dn^{-1/2} \alpha(p, q, n) \text{ i.o.} \right) = 0.$$

Note: The prediction rate for $\hat{\mathbf{f}}$ is **NOT minimax**, because the elastic-net type estimators are **biased**.

Minimax prediction based on a two-stage procedure

The prediction performance can be improved based on a two-stage procedure:

- Stage 1: estimate the signal set $\hat{\mathcal{S}}$
- Stage 2: fit a linear regression model based only on predictors in $\hat{\mathcal{S}}$ using the ℓ_2 penalty:

$$\hat{\mathbf{f}}_{\hat{\mathcal{S}}} = \operatorname{argmin}_{\mathbf{f}_j \in \mathbb{L}_2} \left\{ \frac{1}{n} \sum_{i=1}^n \left(Y_i - \sum_{j \in \hat{\mathcal{S}}} \langle \tilde{X}_{ij}, \mathbf{f}_j \rangle_2 \right)^2 + \lambda \sum_{j \in \hat{\mathcal{S}}} \|\mathbf{f}_j\|_2^2 \right\}.$$

Improvements on the excessive risk based on this procedure can be seen both numerically and theoretically.

To see the theoretical performance, assume that $\hat{\mathcal{S}}$ achieves the variable selection consistency and consider the following simple situation.

Assume that the k -th eigenvalues of $\mathcal{T}^{(j,j)}$ are bounded by ck^{-2r_j} , where $c < \infty$ and $r_j \geq r > 1/2$. This is essentially the model considered in Cai and Yuan (2012) with a single predictor.

Theorem 3

Let $\mathcal{P}(r)$ be a certain class of covariance operators for the true predictors. Then we have the following results:

$$\lim_{a \rightarrow 0} \liminf_{n \rightarrow \infty} \sup_{\widehat{\mathbf{f}} \in \mathcal{P}(r)} \sup_{\mathbf{f}_{0S} \in \mathbb{L}_2^q} \mathbb{P} \left(\mathcal{R}(\widehat{\mathbf{f}}) \geq a(n/q)^{-\frac{2r}{2r+1}} \right) = 1,$$

where the infimum is taken over all possible estimators $\widehat{\mathbf{f}}$.

Theorem 4

Additionally, if the number of true signals q is $o\left(n^{\frac{2r-1}{4r}}\right)$, then

$$\lim_{A \rightarrow \infty} \liminf_{n \rightarrow \infty} \sup_{\widehat{\mathbf{f}} \in \mathcal{P}(r)} \sup_{\mathbf{f}_{0S} \in \mathbb{L}_2^q} \mathbb{P} \left(\mathcal{R}(\widehat{\mathbf{f}}) \geq A(n/q)^{-\frac{2r}{2r+1}} \right) = 0,$$

provided that $\lambda \asymp (n/q)^{-2r/(2r+1)}$.

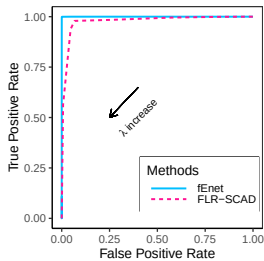
Simulation settings

- ① Functional predictor: $X_{ij}(t) = \sqrt{2} \sum_{k \geq 1} z_{ijk} \sqrt{\nu_k} \cos(k\pi t)$.
- ② $(z_{i1k}, \dots, z_{ipk})^\top \stackrel{i.i.d.}{\sim} N(\mathbf{0}, \Sigma_\rho)$, $\Sigma_\rho = [\rho^{|k-l|}]_{1 \leq k, l \leq p}$.
- ③ Functional coefficients: $\beta_{0j}(t) = 4I(j \leq q) \cdot \sum_{k > 1} (-1)^{u_{jk}} r_k \phi_k(t)$,
 $u_{jk} \stackrel{i.i.d.}{\sim} \text{Bernoulli}(0.5)$.
- ④ $\phi_k(t) = \sqrt{2} \cos(k\pi t)$, and $\nu_k = r_k = \exp(-k/4)$, for $k \geq 1$.
- ⑤ Reproducing kernel: $K(s, t) = -\frac{1}{3} \left\{ B_4(|s - t|/2) + B_4(|s + t|/2) \right\}$,
where $B_k(\cdot)$ is the k th Bernoulli polynomial.

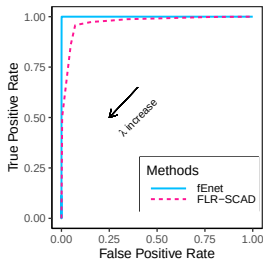
Table: Summary of estimation, prediction, and variable selection performance of the proposed fEnet method versus FLR-SCAD under different problem sizes.

n	p	q	Method	FPR (%)	FNR (%)	MND	RER
$\rho = 0$							
500	50	5	fEnet	0 (0, 0)	0 (0, 0)	0.36 (0.30, 0.45)	0.0006 (0.0003, 0.0009)
			FLR-SCAD	0 (0, 0)	0 (0, 0)	0.54 (0.37, 0.82)	0.0009 (0.0005, 0.0019)
200	100	5	fEnet	0 (0, 0)	0 (0, 0)	0.53 (0.42, 0.68)	0.0018 (0.0011, 0.0029)
			FLR-SCAD	0 (0, 0)	0 (0, 0)	0.75 (0.58, 1.19)	0.0035 (0.0017, 0.0106)
100	200	10	fEnet	0 (0, 1.1)	0 (0, 0)	1.31 (1.06, 1.65)	0.0179 (0.0094, 0.0399)
			FLR-SCAD	4.7 (1.6, 8.4)	0 (0, 30)	4.89 (3.97, 5.00)	0.5280 (0.3206, 0.7734)
$\rho = 0.3$							
500	50	5	fEnet	0 (0, 0)	0 (0, 0)	0.37 (0.31, 0.47)	0.0007 (0.0004, 0.0011)
			FLR-SCAD	0 (0, 0)	0 (0, 0)	0.59 (0.41, 1.03)	0.0012 (0.0006, 0.0027)
200	100	5	fEnet	0 (0, 0)	0 (0, 0)	0.58 (0.45, 0.73)	0.0025 (0.0015, 0.0044)
			FLR-SCAD	0 (0, 0)	0 (0, 0)	0.78 (0.58, 1.51)	0.0044 (0.0021, 0.0146)
100	200	10	fEnet	0 (0, 1.6)	0 (0, 0)	1.39 (1.08, 1.92)	0.0192 (0.0103, 0.0441)
			FLR-SCAD	4.7 (1.6, 9.5)	10 (0, 40)	5.00 (4.37, 5.05)	0.5319 (0.3665, 0.7523)
$\rho = 0.75$							
500	50	5	fEnet	0 (0, 0)	0 (0, 0)	0.53 (0.42, 0.67)	0.0012 (0.0007, 0.0019)
			FLR-SCAD	0 (0, 0)	0 (0, 0)	0.98 (0.67, 1.78)	0.0018 (0.0008, 0.0049)
200	100	5	fEnet	0 (0, 0)	0 (0, 0)	0.85 (0.72, 1.03)	0.0035 (0.0021, 0.0056)
			FLR-SCAD	0 (0, 0)	0 (0, 0)	1.28 (0.76, 4.61)	0.0066 (0.0029, 0.1287)
100	200	10	fEnet	0 (0, 4.2)	0 (0, 10)	2.04 (1.49, 5.00)	0.0175 (0.0078, 0.1329)
			FLR-SCAD	2.1 (0, 4.2)	50 (30, 70)	5.86 (5.00, 7.91)	0.2895 (0.1932, 0.3894)

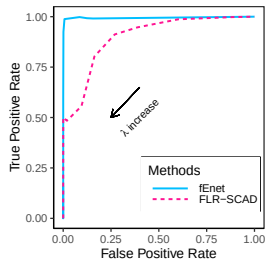
Reparameterize $\lambda_1 = \alpha\lambda$, $\lambda_2 = (1 - \alpha)\lambda$.



(a) $\rho = 0$

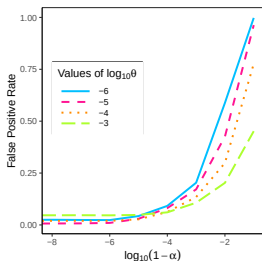


(b) $\rho = 0.3$

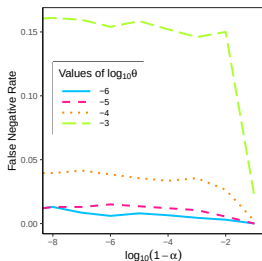


(c) $\rho = 0.75$

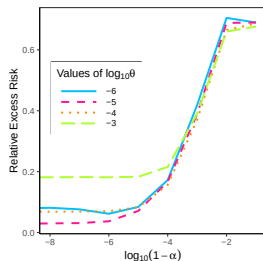
Figure: The ROC curves of fEnet and FLR-SCAD under ultra-high dimension setting $(n, p, q) = (100, 200, 10)$. The ROC curves are obtained by changing the value of λ and holding other hyperparameters at optimal.



(a) FPR



(b) FNR



(c) RER

Figure: The plots of FPR, FNR, and RER versus $\log_{10}(1 - \alpha)$ for different values of θ under ultra high-dimensional case and $\rho = 0.75$.

Table: Relative efficiency (RE) between the functional elastic-net estimate and the two-stage estimate

n	p	q	$\rho = 0$	$\rho = 0.3$	$\rho = 0.75$
500	50	5	1.04	1.06	1.29
200	100	5	1.30	1.44	1.51
100	200	10	1.63	1.68	1.95

Thank You!

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